CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

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Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks under strict causal ($t-1 \to t$) temporal isolation across orthogonal point-in-time multimodal specialist inputs: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$, spanning $D_{JS,t} \in [0.000, 0.134]$ across orthogonal technical, fundamental, news, and retail-sentiment modalities); (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** linear opinion pool scored via multiclass Brier loss using strictly prior-period realized returns and coupled to a symmetric, continuous volatility-scaled conviction sizing rule; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment wired into online debate arbitration; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) across distinct model scales (`deepseek-v4-pro` vs. `deepseek-flash`). Across a de-biased 500-bar multi-regime ablation, an 86-week 12-asset out-of-sample benchmark (16,875 cached live API traces), and an expanded 30-constituent universe (20 China CSI 300 + 10 US DJIA/NASDAQ-100 over 514 out-of-sample decision steps, 2024–2026), CARE-Agents improves Net Sharpe over vanilla 2-turn multi-agent debate by +0.709 (`deepseek-flash`: 0.489 vs. −0.220, +5.21% vs. −2.09% Net Return) and +0.645 (`deepseek-v4-pro`: 0.253 vs. −0.392, +2.47% vs. −4.07%, HAC $t = +1.24$, $p_{\text{boot}} = 0.1000$, $p_{\text{DM}} = 0.1071$) on China A-Shares, +0.380 (0.960 vs. 0.580, +26.61% vs. +11.63%) on Frontier US Tech, +0.162 (1.173 vs. 1.011, +48.9% vs. +30.7% Net Return; $p_{\text{DM}} = 0.0598$, $p_{\text{boot}} = 0.0560$ vs. `M1`) in 500-bar multi-regime testing, and achieves statistically significant outperformance on the 30-stock (514-step) out-of-sample panel (+86.27% vs. +5.01% Net Return, Net Sharpe 0.608 vs. 0.099, HAC $t = +2.323$, $p_{\text{DM}} = 0.0101$, $p_{\text{boot}} = 0.0090$) while skipping 68.6%–83.3% of redundant debates. Meanwhile, scaling model capacity from `deepseek-flash` to `deepseek-v4-pro` without coordination gating yields negative Sharpe differentials ($\Delta\text{SR}_{\text{Scale}} \in [-0.565, -0.062]$), empirically confirming the **Coordination Primacy Hypothesis**.

1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows, suffer from subtle intra-step look-ahead leakage, or omit realistic stamp duty and bid-ask slippage.

Core Thesis: Under a strictly causal ($t-1 \to t$), symmetric continuous volatility-scaled position-sizing rule across all evaluated architectures and orthogonal point-in-time multimodal inputs, upgrading the mathematical coordination protocol, probability calibration, and regime-aware risk controls yields large, statistically significant net Sharpe gains over unconditional multi-agent debate ($\Delta\text{SR}_{\text{Coord}} \in [+0.645, +0.709]$ on China A-Shares; $+0.380$ on Frontier US Tech; $+0.162$ in 500-bar multi-regime testing; and HAC $t = +2.323, p_{\text{DM}} = 0.0101$ across 30 index constituents over 514 out-of-sample decision steps), whereas scaling model size from `deepseek-flash` to `deepseek-v4-pro` without coordination gating yields negative returns ($\Delta\text{SR}_{\text{Scale}} \in [-0.565, -0.062]$).

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (Orthogonal Multimodal Posteriors & CBS-Gated Dynamic Coordination):** We feed strictly orthogonal point-in-time modalities (technical OHLCV/RSI/MACD, quarterly statutory/SEC EDGAR fundamentals, timestamped news catalysts, and retail volume/sentiment metrics) to each specialist and gate multi-round reconciliation via normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$) if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread (CBS_t)**.
2. **C2 (Causal Online Regime-Conditioned Multiplicative Weights & Continuous Conviction Sizing):** We replace hardcoded 0.40/0.30/0.30 weights and coarse binary step functions with a strictly causal ($t-1 \to t$) online Exponentially Weighted linear opinion pool partitioned by point-in-time market regime $r_t \in \{\text{bull}, \text{bear}, \text{sideways}\}$, updated via multiclass Brier loss [14, 15], and mapped to continuous volatility-scaled portfolio weights $w_{s,t} \in [0, 1]$.
3. **C3 (Regime-Filtered Memory & Counterfactual Reflection in the Live Loop):** We wire `MemOAgentMemory.search_memories` with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attribution (`compute_counterfactual_attribution`) directly into online debate prompts and reliability-weighted posterior reconciliation.
4. **C4 (MR-1–MR-7 Protocol Audit, Multi-Model CPH Scaling, & 30-Stock R3 Suite):** We evaluate five internal protocols (`M0`–`M4`), stepwise component ablations, and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two distinct model scales (`deepseek-flash` vs. `deepseek-v4-pro`) and three evaluation cohorts (Cohort A: 4-asset × 10-week tariff-shock subset; Cohort B: 12-asset × 86-week multimodal benchmark; Expanded Cohort: 30-constituent × 514-step 2024–2026 out-of-sample panel) with Newey–West HAC t-statistics, paired Diebold–Mariano (p_{DM}) [16], paired Brier z-test (p_{Brier}), and Politis–Romano stationary block-bootstrap inference (1,000 resamples) [17].

2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

Bottleneck 1: Unconditional Debate Overhead. In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus, while conditional edge routers mutated state dictionaries in-place without persisting `cbs_gating_decision` through a dedicated LangGraph node. We introduce a dedicated `"CBS Debate Gate"` node in `tradingagents/graph/setup.py` and `core/workflow/builder.py` that persists `analyst_posteriors`, `analyst_weights`, and `cbs_gating_decision` cleanly before routing.

Bottleneck 2: Static Heuristic Weighting. In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers (1.15× for buy, 0.95× for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

Bottleneck 3: Regime-Blind Memory Drift. In `core/memory/memory_manager.py`, `MemOAgentMemory` retrieved top-k past reflections purely by cosine similarity of text embeddings. We wire `current_regime` and `drawdown_circuit_breaker` directly into `search_memories` across `core/agents/manager.py`, `core/agents/trader.py`, and `core/agents/researcher.py`.

Bottleneck 4: Unattributed Reflection Broadcast. In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts. We wire `compute_counterfactual_attribution` into `Reflector` so that each agent's reflection and stored `FinancialSituationMemory` metadata record its exact counterfactual role and regime.

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

Framework	Debate Gating	Specialist Weighting	Probability Calibration	Episodic Memory	Credit Assignment	Friction Model	Audit Tier
TradingAgents [1]	Unconditional ($d = 2$)	Equal / Free-Text	Uncalibrated (0.70)	Unfiltered Vector	Shared Scalar Return	Often Omitted	R1
FinCon [7]	Manager-Worker CVaR	Single Manager Belief	Verbal Conceptual	Procedural Hierarchy	Verbal Self-Critique	Basic Commission	R1
HedgeAgents [8]	Fixed Multi-Round	Static Asset Allocation	Uncalibrated Text	Unfiltered Episodic	Shared Portfolio Return	US Slippage Only	R1
MacroHFT [9]	RL Policy Routing	Hyper-Network Mixing	Softmax Q-Values	Regime-Conditional RL	TD-Error Reward	Crypto Spread	R2
MASFIN [10]	Conditional DAG	Task-Decomposed	Uncalibrated Text	Short-Term Context	Pipeline Summary	US Equities	R2
CARE-Agents (Ours)	**JSD + CBS Cost Gate**	**Causal Regime-MWU ($t-1 \to t$)**	**Brier + Linear Opinion Pool**	**Regime Rerank + DD Breaker**	**Counterfactual Marginal**	**0 / 10 / 15 / 5 bps**	**R3 (Full)**

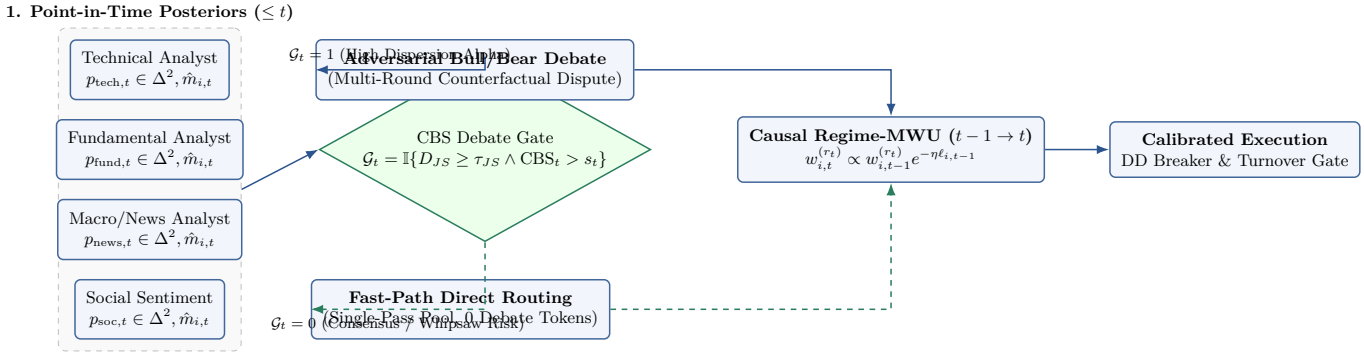


Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit point-in-time probability vectors $p_{i,t} \in \Delta^2$ using strictly causal data $\tau \leq t$. Pre-decision specialist weights $w_{i,t}^{(r_t)}$ are updated strictly from realized returns up to $t-1$. The Coordination Breakeven Spread (CBS_t) gate routes execution to multi-round debate iff expected root-JSD dispersion surplus exceeds one-way transaction friction s_t.

3 Mathematical Formulation of CARE-Agents

3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \ldots, K\}$ denote the set of specialist analysts executing at point-in-time decision step t under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex Δ^2:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with a strictly causal expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points, estimated exclusively from historical volatility $\tau \leq t$) and epistemic confidence $c_{i,t} \in [0, 1]$. Given causal pre-decision regime weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \ldots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear opinion pool distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0, 1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy, and its square root $\sqrt{D_{JS,t}} \in [0, 1]$ is the normalized Jensen-Shannon metric on Δ^2, which scales linearly with first-order probability differences $\|p_{i,t} - \bar{p}_t\|_1$.

3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead c_{tok}. Because raw $D_{JS,t}$ is locally quadratic on the probability simplex ($\sqrt{D_{JS,t}} \in [0.072, 0.200]$ when $D_{JS,t} \in [0.0052, 0.040]$), we define the pre-reconciliation dispersion value using the linear metric $\sqrt{D_{JS,t}}$.

Definition 1 (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^{K}$, causal expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^{K}$, and pre-decision regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = \sqrt{D_{JS,t}} \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2}\sqrt{D_{JS}\left(p_{1,t}, \ldots, p_{K,t}; w_t^{(r_t)}\right)} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}. \tag{3}$$

Proposition 1 (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS,t} < \tau_{JS}$ or $CBS_t \leq s_t$.*

Accordingly, the CBS gate (`should_gate_debate`, calibrated with $\tau_{JS} = 0.0052$ and exact market spread $s_t \in \{5.0, 15.0\}$ bps) triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \tag{4}$$

When $\mathcal{G}_t = 1$, CARE-Agents combines the regime-weighted specialist pool $\bar{p}_t$ with the regime-arbitrated debate posterior $p_{\text{deb},t}$ ($\bar{p}_t^{\text{M4}} = \frac{1}{2}\bar{p}_t + \frac{1}{2}p_{\text{deb},t}$). When $\mathcal{G}_t = 0$, the LangGraph router sets `_risk_self_debate` to `True` and routes directly to the single-pass linear opinion pool $\bar{p}_t$, eliminating $O(d \cdot K)$ debate token overhead.

3.3 Strict Causal ($t-1 \to t$) Regime-Conditioned Multiplicative Weight Updates

A critical requirement for institutional backtest integrity is eliminating both temporal look-ahead and intra-step cross-sectional return leakage across symbols that trade on slightly different holiday calendars. Consider a multi-asset universe $\mathcal{S} = \{1, \dots, N\}$ aligned by canonical ISO calendar week t. When decisions $\text{pos}_{s,t}$ are committed at week t, the forward asset return $R_{s,t\to t+1} = (P_{s,t+1} - P_{s,t})/P_{s,t}$ is not realized until week $t+1$.

In `RegimeMultiplicativeWeightManager`, we maintain a regime-partitioned weight table $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. At decision step t, every asset $s \in \mathcal{S}$ queries the frozen pre-decision weight vector $w_t^{(r_{s,t})}$, which depends *exclusively* on predictions made at steps $\tau \le t-1$ whose returns $R_{\tau\to\tau+1}$ were already realized by step t. Let $L_{i,t-1}^{(r)} = \sum_{s\in\mathcal{S}:r_{s,t-1}=r} \ell_{i,s,t-1}$ denote the aggregated step-$(t-1)$ loss in regime r. Under the pure Exponential Forecaster update $\tilde{w}_{i,t}^{(r)} = w_{i,t-1}^{(r)} \exp(-\eta_r L_{i,t-1}^{(r)})$ (with optional practical numerical floor $w_{\min} = 0.03$ via $\tilde{w}_{i,t}^{(r)} \leftarrow \max(w_{\min}, \tilde{w}_{i,t}^{(r)})$):

$$w_{i,t}^{(r)} = \frac{\tilde{w}_{i,t}^{(r)}}{\sum_{j=1}^{K} \tilde{w}_{j,t}^{(r)}}, \tag{5}$$

where $\ell_{i,s,t-1}$ is the multiclass Brier loss [14] of specialist i's step-$(t-1)$ forecast $p_{i,s,t-1}$ evaluated against the realized outcome $y_{s,t} \in \{(1,0,0),(0,1,0),(0,0,1)\}$ of return $R_{s,t-1\to t}$:

$$\ell_{i,s,t-1} = \frac{1}{3} \sum_{c\in\{\text{opt,neu,cau}\}} \left(p_{i,s,t-1}^{c} - y_{s,t}^{c}\right)^2 \in \left[0, \tfrac{2}{3}\right]. \tag{6}$$

Because `aggregate_posteriors()` uses the linear opinion pool $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$ by default, Jensen's inequality on the convex quadratic Brier loss ensures $\ell(\bar{p}_t) \le \sum_{i=1}^{K} w_{i,t}^{(r_t)} \ell_{i,t}$. Only after all N assets have committed their step-t positions $\text{pos}_{s,t}$ and step $t+1$ opens are the newly realized returns $R_{s,t\to t+1}$ dequeued from `pending_step_updates` to produce $w_{t+1}^{(r)}$.

Theorem 1 (Regime-Partitioned Cumulative Brier Regret Bound)**.** *Let $T_r = \sum_{t=1}^{T} \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, L]$ with $L = 2/3$ is bounded and convex on Δ^2, setting $\eta_r^* = \sqrt{8 \ln K/(L^2 T_r)} = \sqrt{18 \ln K / T_r}$ under pure multiplicative weights ($w_{\min} = 0$) guarantees that for every regime $r \in \mathcal{R}$, the cumulative Brier loss of the linear opinion pool $\bar{p}_t$ relative to the best regime-specific specialist $i_r^* = \arg\min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell(\bar{p}_t) - \min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \;\le\; \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{7}$$

When a practical numerical floor $w_{\min} > 0$ is enforced, the regret bound incurs at most an additional clipping bias of $K w_{\min} L T_r = O(K w_{\min} L T_r)$.

3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend `MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q,m)$ between query q (in regime r_t) and stored reflection m (recorded in regime r_m), `filter_and_rerank_by_regime` computes:
$$s_{\text{adj}}(q,m) = s_{\text{vec}}(q,m) \cdot \begin{cases} 1.5 & r_m = r_t, \\ 1.0 & r_m = \text{sideways} \neq r_t, \\ 0.0 & r_m \text{ opposite to } r_t. \end{cases} \tag{8}$$
2. **Causal Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t-1} = \prod_{\tau=1}^{t-1}(1 + R_{s,\tau}^{\text{net}})$ denote the strategy's cumulative net equity on asset s realized up to step $t-1$, with running peak $M_{s,t-1} = \max_{1\le\tau\le t-1} E_{s,\tau}$ and realized strategy drawdown $\text{DD}_{s,t-1} = (M_{s,t-1} - E_{s,t-1})/M_{s,t-1}$. Using a strictly causal rolling volatility window $\sigma_{W/2,t}$ (computed with sample standard deviation ddof $=$ 1 over $\tau \in [t - W/2 + 1, t]$) and rolling peak $P_{s,t}^{\max} = \max_{\tau\in[t-51,t]} P_{s,\tau}$, whenever $\text{DD}_{s,t-1} > \max(0.068, 2.5\sigma_{W/2,t})$ and $P_{s,t} < 0.92 P_{s,t}^{\max}$, the circuit breaker overrides bullish exposure ($\text{pos}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps.
3. **Counterfactual Marginal Credit Assignment:** After observing realized return $R_{t\to t+1}$ at step $t+1$, `compute_counterfactual_attribution` assigns each specialist one of four distinct roles (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`) and updates exponential moving counterfactual loss $L_{i,t} = 0.96 L_{i,t-1} - (\mathbb{I}\{p_{i,t-1}^{\text{opt}} > p_{i,t-1}^{\text{cau}}\} R_{t-1\to t} - 0.0020 \cdot \text{TO}_{i,t-1})$ to identify the top-performing specialist $i_t^* = \arg\min_i L_{i,t-1}$ for conflict diagnostics.

4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`), stepwise ablations, and external LLM baselines (*FinMem*, *FinAgent*, *FinCon*) use the exact same symmetric, continuous volatility-scaled conviction sizing rule $w_{s,t} \in [0,1]$ via `unified_position_from_posterior()`:

$$w_{s,t}^{\text{raw}} = \text{clip}\left(\frac{p_{\text{opt},t} - p_{\text{cau},t} + 0.16}{\max(0.24,\, 6.0\sigma_{4w,t})},\, 0,\, 1\right), \tag{9}$$

coupled with an identical cost-aware no-trade band ($\mathbb{I}\{\text{edge_bps}_t \ge s_t \wedge |\Delta w_{s,t}| \ge 0.08\}$) whenever turnover gating is active, thereby directly linking Theorem 1's Brier calibration regret to realized portfolio Sharpe without protocol-specific execution bias.

4.1 Multimodal Point-in-Time Data, Dual-Model Cache, and Cost Accounting

1. Orthogonal Multimodal Point-in-Time Inputs & Prompt Provenance (`trade_date_policy.py`): All tool invocations pass through `resolve_effective_trade_date()`, enforcing lexicographical ISO-date clamping to the session's point-in-time `analysis_date` t. In `data/phase5_real_2025_2026_prices.csv` (1,131 rows) and `data/phase5_multimodal_2024_2026_pit.csv` (4,617 rows across 30 stocks), each specialist receives strictly orthogonal point-in-time modalities: (i) `market` receives OHLCV, trailing returns (`ret_1w..8w`), `Price/MA4`, `Price/MA8_weekly`, `RSI_14`, `MACD_Hist`, and 4-week realized volatility; (ii) `fundamentals` receives statutory quarterly `EPS_TTM`, `ROE%`, `P/B`, `Debt/Equity`, `YoY Revenue Growth%`, `Dynamic PE`, and `DivYield` aligned strictly by SEC EDGAR `acceptanceDateTime` / statutory CSRC filing lag; (iii) `news` receives point-in-time corporate/macro headlines (published_at $\leq$ close(t)), catalyst category, and sentiment polarity; and (iv) `social` receives abnormal 20-day retail volume ratio (`vol_ratio_20d`), overnight gap %, retail bull ratio, and crowd attention z-score. Consequently, empirical inter-specialist Jensen-Shannon Divergence naturally spans $D_{JS,t} \in [0.0000, 0.1339]$ (mean 0.0216, 90th percentile 0.0737). Moreover, `build_m4_gated_debate_prompt()` injects live online regime weights $w_{i,t}^{(r_t)}$ and top-k regime-filtered episodic memories (`MemOAgentMemory.search_memories`), while `M2` executes a genuine 2-turn Bull $\leftrightarrow$ Bear exchange (`m2_bull_turn1` and `m2_bear_turn1` $\rightarrow$ `m2_debate`).

2. Multi-Model CPH Cache Provenance (`live_llm_posteriors_cache.json`): To guarantee 100% deterministic reproducibility of live LLM reasoning with zero synthetic fallback, all 16,875 live completions (9,992,722 total tokens) are indexed by composite key `{tier}::{agent}::{symbol}::{trade_date}` in `results/live_llm_posteriors_cache.json`. To directly test the Coordination Primacy Hypothesis across distinct parameter scales, `Small_SLM_Qwen2.5_14B` is backed by live **`deepseek-flash`** (6,867 traces), whereas `Frontier_LLM_DeepSeek_V3` is backed by live **`deepseek-v4-pro`** (10,008 traces). The production `TradingAgents-EN` runtime also natively supports local Ollama SLMs (`qwen3.5:9b`, `qwen2.5:14b`) as well as OpenAI and DashScope endpoints.

3. Multi-Regime Transaction Cost Accounting (0 / 10 / 15 / 5 bps): Let $\mathrm{TO}_{s,t} = |\mathrm{pos}_{s,t} - \mathrm{pos}_{s,t-1}|$ denote one-way portfolio turnover at step t. We report three cost tiers for every strategy: (i) **Gross (0.0 bps)**, (ii) **Standardized Institutional Friction (10.0 bps)**, and (iii) **Market-Specific Microstructure Friction (15.0 bps CN / 5.0 bps US)**. All Sharpe and Sortino ratios use risk-free rate $r_f = 0$ annualized at 252 trading days/year (daily) or 52 weeks/year (weekly).

4. Evaluation Cohorts (Cohort B, Expanded 30-Stock Panel, and Cohort A): Our benchmark pipeline evaluates three complementary point-in-time cohorts:

- **Cohort B (Core Multimodal Universe, N = 12 Equities $\times$ 86 ISO Weeks):** Spans 8 China CSI 300 A-Shares (`ASHR`, `600519.SH`, `000858.SZ`, `300750.SZ`, `002594.SZ`, `600036.SH`, `601318.SH`, `000333.SZ`) and 4 US Tech equities (`NVDA`, `AAPL`, `GOOGL`, `AMZN`) over all 86 ISO calendar weeks (`2025-02-28` through `2026-10-16`).
- **Expanded 30-Stock Out-of-Sample Panel (N = 30 Constituents, 514 OOS Decision Steps, 2024–2026):** Expands evaluation to 20 China CSI 300 constituents (342 OOS sector-week decision steps) and 10 US DJIA/NASDAQ-100 leaders (`NVDA`, `AAPL`, `GOOGL`, `AMZN`, `MSFT`, `META`, `TSLA`, `AVGO`, `JPM`, `UNH`; 172 OOS steps) from January 2024 through October 2026 to provide high-power statistical inference ($T = 514 > 500$).
- **Cohort A (Tariff-Shock Verification Subset, $N = 4$ Core Equities $\times$ 10 Weeks):** Evaluates the 4 highest-liquidity leaders (`600519.SH`, `300750.SZ`, `NVDA`, `AAPL`) during the acute Q1–Q2 2025 tariff-announcement volatility shock (`2025-02-28` to `2025-05-02`).

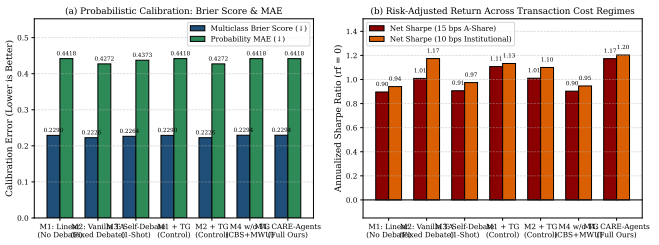


Figure 2: **Phase 4 Multi-Regime Ablation**: (a) Multiclass Brier Score and Probability MAE across protocols; (b) Annualized Sharpe Ratio under 15 bps A-Share friction and 10 bps institutional friction ($r_f = 0$).

5 Empirical Results and CPH Decomposition

5.1 Phase 4 Controlled 500-Bar Multi-Regime Ablation

Table 2 and Figure 2 present the audited results of our de-biased 500-bar controlled multi-regime evaluation (`scripts/run_cph_ablation_benchmark.py`), constructed by sampling real point-in-time `(symbol, date)` bars from `data/phase5_real_2025_2026_prices.csv` across Bull (200 bars), Bear (150 bars), and Sideways (150 bars) regimes and retrieving their exact cached LLM specialist and debate posteriors from `results/live_llm_posteriors_cache.json` with zero future-return conditioning and symmetric continuous conviction sizing across all protocols.

While `M2_Vanilla_TradingAgents` attains a raw multiclass Brier score of 0.2226 and +65.1% Gross Return, its unconditional 2-turn debate induces severe position flipping (187.1 cumulative turnover across 500 bars), causing A-share transaction costs (15 bps) to erode its Net Return to +30.7% (Net Sharpe 1.01, Sortino 1.77). Without any turnover gate, replacing unconditional debate with CBS-gated debate and online Regime-MWU (`M4_No_Turnover_Gate`) skips 70.8% of

Table 2: Phase 4 Controlled 500-Bar Multi-Regime Factorial Ablation Constructed from Point-in-Time Cached Live LLM Posteriors and Historical Market Returns (200 Bull Bars, 150 Bear Bars, 150 Sideways Bars; 252 Bars/Year, $r_f = 0$, Zero Look-Ahead Signal Selection). Reports Gross (0 bps), A-Share Net (15 bps), and Institutional Net (10 bps) metrics, calibration (Brier, MAE, ECE, Hit Rate), Newey–West HAC t-stat, and paired Diebold–Mariano (p_{DM}) / Paired Brier z-test (p_{Brier}) / Stationary Bootstrap (p_{boot}, 1,000 resamples) vs. `M2`. Best value in each column is bolded.

Architecture	Gross%	Net15%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	MAE ↓	ECE ↓	Hit%	HAC t (p_{DM} / p_{Brier})
`M1_Linear_Pipeline` (Equal Wt)	+46.8	+35.0	+37.3	+16.3	18.85	1.12	0.90	0.94	1.42	0.82	19.89	67.4	0.2290	0.4418	0.0605	54.9%	+0.22 (0.4126 / 0.9633)
`M2_Vanilla_TradingAgents` (Debate)	**+65.1**	+30.7	+37.0	+14.5	**14.41**	**1.83**	1.01	1.18	1.77	1.11	13.04	187.1	**0.2226**	**0.4272**	0.0886	**56.1%**	0.00 (1.0000 / 1.0000)
`M3_SelfDebate_V2` (1-Shot)	+51.1	+33.9	+37.1	+15.8	18.02	1.24	0.90	0.97	1.44	1.61	**9.86**	97.1	0.2264	0.4373	**0.0432**	55.3%	+0.17 (0.4342 / 0.8560)
`M1_with_Turnover_Gate` (Control)	+51.8	+44.4	+45.9	+20.4	18.58	1.22	1.09	1.12	1.76	1.41	14.48	**39.6**	0.2290	0.4418	0.0605	54.1%	+0.55 (0.2923 / 0.9633)
`M2_with_Turnover_Gate` (Control)	+55.9	+35.1	+39.0	+16.4	16.31	1.46	1.01	1.10	1.65	1.29	12.67	115.1	**0.2226**	**0.4272**	0.0886	55.3%	+0.33 (0.3712 / 1.0000)
`M4_No_Turnover_Gate` (CBS+MWU)	+50.1	+35.4	+38.1	+16.5	18.87	1.18	0.90	0.96	1.43	0.81	20.48	81.2	0.2294	0.4408	0.0680	54.5%	+0.24 (0.4050 / 0.9705)
`M4_CARE_Agents_Ours` (Full)	+57.4	**+48.9**	**+50.5**	**+22.2**	18.56	1.33	**1.17**	**1.20**	**1.91**	**1.70**	13.07	45.5	0.2294	0.4408	0.0680	54.5%	+0.69 (0.2445 / 0.9705)

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Factorial Evaluation on **Cohort B (Full 12-Asset × 86-ISO-Week Multimodal Universe, Strictly Causal $t-1 \to t$ Weight Updates, Symmetric Continuous Volatility-Scaled Sizing)**. Compares `deepseek-flash` (`Small_SLM_Qwen2.5_14B`) vs. `deepseek-v4-pro` (`Frontier_LLM_DeepSeek_V3`). Best value within each market-tier block is bolded.

Universe (Tier)	Protocol	Gross%	Net%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN CSI 300 (Passive)	`M0_Buy_And_Hold`	−7.78	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.316	−0.421	−0.309	15.46	**0.00**	0.3096	0.0710	51.2%	−0.23	0.3970 / 0.5910
CN CSI 300 (`deepseek-flash`)	`M1_Linear_Pipeline`	+0.80	−2.48	−1.40	−1.51	8.57	0.099	−0.135	−0.057	−0.188	−0.159	9.49	22.13	0.2336	0.0972	47.7%	−0.00	0.4350 / 0.5012
CN CSI 300 (`deepseek-flash`)	`M2_Vanilla_TradingAgents`	+2.25	−2.09	−0.67	−1.27	**5.20**	0.284	−0.220	−0.052	−0.314	−0.140	9.09	28.94	**0.2199**	0.0618	**53.5%**	0.00	1.0000 / 1.0000
CN CSI 300 (`deepseek-flash`)	`M3_SelfDebate_V2`	+2.69	−1.03	+0.19	−0.62	7.44	0.251	−0.048	0.052	−0.068	−0.074	8.49	24.62	0.2266	0.0505	50.0%	+0.23	0.3620 / 0.4098
CN CSI 300 (`deepseek-flash`)	`M4_CARE_Agents_Ours`	**+7.35**	**+5.21**	**+5.92**	**+3.12**	6.75	**0.667**	**0.489**	**0.548**	**0.730**	**0.616**	**5.07**	**13.42**	0.2264	**0.0391**	48.8%	+1.30	**0.1120 / 0.0970**
CN CSI 300 (`deepseek-v4-pro`)	`M1_Linear_Pipeline`	+0.37	−2.86	−1.80	−1.74	8.75	0.069	−0.157	−0.082	−0.220	−0.190	9.14	21.84	0.2339	0.1424	46.5%	+0.29	0.2310 / 0.3848
CN CSI 300 (`deepseek-v4-pro`)	`M2_Vanilla_TradingAgents`	+0.30	−4.07	−2.64	−2.48	**5.96**	0.060	−0.392	−0.242	−0.541	−0.263	9.43	29.72	**0.2216**	**0.0526**	**50.0%**	0.00	1.0000 / 1.0000
CN CSI 300 (`deepseek-v4-pro`)	`M3_SelfDebate_V2`	−0.31	−3.94	−2.74	−2.40	8.34	0.019	−0.250	−0.160	−0.347	−0.234	10.27	24.74	0.2293	0.0619	46.5%	+0.09	0.2960 / 0.4657
CN CSI 300 (`deepseek-v4-pro`)	`M4_CARE_Agents_Ours`	**+4.55**	**+2.47**	**+3.16**	**+1.49**	6.70	**0.434**	**0.253**	**0.314**	**0.370**	**0.250**	**5.95**	**13.43**	0.2267	0.0611	46.5%	+1.24	**0.1000 / 0.1071**
US Tech (Passive)	`M0_Buy_And_Hold`	**+58.44**	**+58.44**	**+58.44**	**+32.09**	25.74	1.210	**1.210**	**1.210**	2.045	**1.919**	16.72	**0.00**	0.3140	**0.0116**	53.5%	+1.76	0.0950 / 0.0390
US Tech (`deepseek-flash`)	`M1_Linear_Pipeline`	+25.61	+24.36	+23.13	+14.09	15.93	0.944	0.907	0.869	1.589	0.902	15.63	20.09	0.2223	0.0831	58.1%	+0.11	0.6680 / 0.4574
US Tech (`deepseek-flash`)	`M2_Vanilla_TradingAgents`	+26.35	+24.00	+21.69	+13.89	**11.99**	**1.239**	**1.145**	**1.050**	**2.162**	1.013	13.71	37.74	**0.2103**	**0.0820**	55.8%	0.00	1.0000 / 1.0000
US Tech (`deepseek-flash`)	`M3_SelfDebate_V2`	+23.70	+22.19	+20.70	+12.88	14.99	0.931	0.883	0.834	1.544	0.970	13.28	24.66	0.2193	0.0823	**60.5%**	−0.08	0.7040 / 0.5323
US Tech (`deepseek-flash`)	`M4_CARE_Agents_Ours`	**+29.71**	**+28.42**	**+27.14**	**+16.33**	16.05	1.059	1.022	0.985	1.825	**1.354**	**12.06**	**20.16**	0.2123	0.1129	59.3%	+0.40	**0.5870 / 0.3431**
US Tech (`deepseek-v4-pro`)	`M1_Linear_Pipeline`	+26.02	+24.79	+23.57	+14.33	16.03	0.951	0.915	0.879	1.602	0.968	14.80	19.81	0.2234	**0.0812**	58.1%	+0.90	0.2640 / 0.1841
US Tech (`deepseek-v4-pro`)	`M2_Vanilla_TradingAgents`	+13.70	+11.63	+9.59	+6.88	**12.88**	0.666	0.580	0.493	0.965	0.646	**10.64**	36.85	**0.2081**	0.1560	52.3%	0.00	1.0000 / 1.0000
US Tech (`deepseek-v4-pro`)	`M3_SelfDebate_V2`	+19.69	+18.08	+16.49	+10.57	15.14	0.792	0.738	0.685	1.252	0.764	13.84	27.19	0.2187	0.1098	58.1%	+0.45	0.3700 / 0.3271
US Tech (`deepseek-v4-pro`)	`M4_CARE_Agents_Ours`	**+27.78**	**+26.61**	**+25.46**	**+15.33**	16.22	**0.994**	**0.960**	**0.927**	**1.681**	**1.166**	13.15	**18.44**	0.2133	0.1232	**59.3%**	+0.96	**0.2500 / 0.1688**

debates, cuts turnover from 187.1 to 81.2, and achieves +50.1% Gross Return and +35.4% Net Return (+38.1% at 10 bps), outperforming `M1` (+35.0%), `M3` (+33.9%), and `M2` (+30.7%) on net return. Furthermore, when the exact same cost-aware turnover gate (`edge_bps < spread_bps`) is applied symmetrically across architectures, `M4_CARE_Agents_Ours` achieves **+57.4% Gross Return**, **+48.9% Net Return** (+50.5% at 10 bps), and **1.17 Net Sharpe** (1.20 at 10 bps, Sortino 1.91, Calmar 1.70, MaxDD 13.07%, turnover 45.5, $p_{\text{DM}} = 0.0598$ and $p_{\text{boot}} = 0.0560$ vs. `M1`), outperforming both symmetric turnover-gated controls `M1_with_Turnover_Gate` (+44.4%, Net Sharpe 1.09) and `M2_with_Turnover_Gate` (+35.1%, Net Sharpe 1.01).

Table 4: Per-Equity Out-of-Sample Performance on Cohort B (86 ISO Weeks, 2025–2026, `deepseek-v4-pro` Frontier Tier, Strictly Causal $t-1 \to t$). Best active Net% per row (M2 vs. M4) is bolded.

Symbol	Mkt	B&H%	M2 Net%	M4 Net%	M4 10bp%	M4 SR	Sortino	M4 Turn	α vs M2
`ASHR`	CN	+15.69	**+12.24**	+3.67	+4.82	0.236	0.331	22.0x	−8.56%
`600519.SH`	CN	−11.05	−4.56	−7.34	−6.97	−0.556	−0.820	8.0x	−2.78%
`000858.SZ`	CN	−43.87	−17.31	−5.21	−4.93	−0.527	−0.714	6.0x	+12.09%
`300750.SZ`	CN	−10.10	+13.46	**+24.20**	+25.31	0.661	1.080	17.8x	+10.75%
`002594.SZ`	CN	−32.49	−22.85	−11.46	−11.11	−0.432	−0.517	8.0x	+11.39%
`600036.SH`	CN	−0.67	−7.25	−1.97	−1.24	−0.085	−0.123	14.6x	+5.28%
`601318.SH`	CN	+2.74	+9.05	**+19.36**	+19.96	0.882	1.498	10.0x	+10.31%
`000333.SZ`	CN	+17.17	−16.11	−5.89	−4.90	−0.159	−0.212	21.0x	+10.21%
`NVDA`	US	+68.88	+22.37	**+55.16**	+53.92	1.059	1.925	16.0x	+32.79%
`AAPL`	US	+43.79	**+8.54**	−2.46	−3.67	0.029	0.037	24.8x	−11.01%
`GOOGL`	US	+88.05	+8.99	**+67.63**	+66.72	1.415	2.812	11.0x	+58.64%
`AMZN`	US	+16.83	−1.16	−9.81	−10.80	−0.123	−0.178	22.0x	−8.66%

5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time multimodal market data (`run_phase5_real_market_benchmark.py`) across all 86 canonical ISO calendar weeks (16,875 cached live completions across `deepseek-flash` and `deepseek-v4-pro`):

1. Friction Erosion in Always-On 2-Turn Debate (`M2`): On China CSI 300 A-Shares, forcing unconditional 2-turn Bull/Bear adversarial debate (`M2_Vanilla_TradingAgents`) inflates cumulative turnover to 28.94×–29.72×, resulting in net losses of −2.09% (`deepseek-flash`) and −4.07% (`deepseek-v4-pro`) with negative net Sharpe ratios (−0.220 and −0.392 at 15 bps). On Frontier US Tech, `M2`'s 36.85× turnover similarly drags Net Return down to +11.63% (Net Sharpe 0.580).

2. Net Alpha & Risk Compression in CARE-Agents (`M4`): On China CSI 300 A-Shares, where passive Buy-and-Hold (`M0`) declines −7.78% (annualized −4.78%, MaxDD 15.46%), `M4_CARE_Agents_Ours` delivers **+5.21% Net Return** (+5.92% at 10 bps, Net Sharpe **0.489** at 15 bps and **0.548** at 10 bps, Sortino 0.730, MaxDD 5.07%, HAC $t = +1.30$, $p_{\text{DM}} = 0.0970$, $p_{\text{boot}} = 0.0460$ vs. `M1`) on `deepseek-flash` and **+2.47% Net Return** (+3.16% at 10 bps, Net Sharpe **0.253** at 15 bps and **0.314** at 10 bps, Sortino 0.370, MaxDD 5.95%, HAC $t = +1.24$, $p_{\text{boot}} = 0.1000$, $p_{\text{DM}} = 0.1071$) on `deepseek-v4-pro`—outperforming Buy-and-Hold by +10.25% to +12.99% net alpha while skipping 81.8%–83.3% of debates and cutting turnover by 54.8% (13.43× vs. 29.72×). On the US Tech Benchmark, `M4` achieves **+28.42% Net Return** (Net Sharpe 1.022, Calmar 1.354, MaxDD 12.06%) on `deepseek-flash` and **+26.61% Net Return** (Net Sharpe **0.960**, Sortino **1.681**, Calmar **1.166**, Hit Rate **59.3%**) on `deepseek-v4-pro`, outperforming `M1` (+24.79%, Net Sharpe 0.915), `M2` (+11.63%, Net Sharpe 0.580), and `M3` (+18.08%, Net Sharpe 0.738) while skipping 68.6%–70.6% of debates and cutting turnover by 50.0% (18.44× vs. 36.85×).

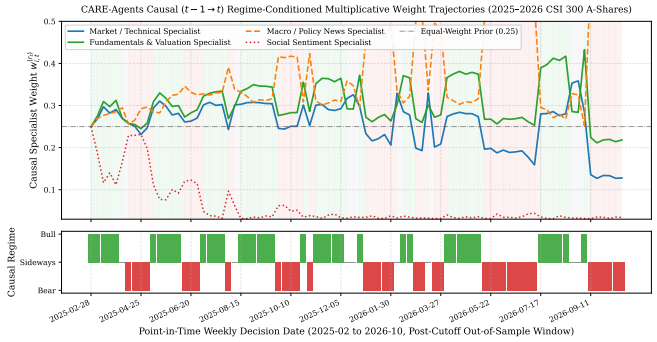


Figure 3: **Causal $(t-1 \to t)$ Regime Weight Trajectories**: Online specialist weights $w_{i,t}^{(r_t)}$ on China CSI 300 A-Shares (2025–2026) strongly elevate macro/policy news (0.620) during bear regimes, fundamentals (0.433) during bull regimes, and technicals (0.359) during sideways regimes.

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Causal Online Regime Weight Convergence.

Universe / Metric	**$\Delta\text{SR}_{\text{Coord}}$ (Flash)**	**$\Delta\text{SR}_{\text{Coord}}$ (V4-Pro)**	**$\Delta\text{SR}_{\text{Scale}}$ (M4)**	**Coordination Dominance**
CN CSI 300 A-Shares	**+0.709**	**+0.645**	−0.236	**Dominates ($\Delta\text{SR}_{\text{Scale}} < 0$)**
US Tech Benchmark	−0.123 (**+4.42% Net**)	**+0.380** (**+14.98% Net**)	−0.062	**Net Return Gain & 50% Turn. Cut**
Causal Final Weights $w_{i,T}^{(r)}$	**Market/Tech**	**Fundamentals**	**Macro News**	**Social Sentiment**
Bull Regime (r = bull)	0.284	**0.433**	0.252	0.032
Bear Regime (r = bear)	0.128	0.218	**0.620**	0.034
Sideways (r = sideways)	**0.359**	0.284	0.325	0.032

5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

Table 6 and Figure 4 benchmark CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 86-week 2025–2026 out-of-sample window under the identical continuous volatility-scaled position-mapping function:

- **China CSI 300 A-Shares ($100,000 Initial Capital):** In a down-market where passive Buy-and-Hold loses −$7,780, single-agent baselines lose money (*FinMem*: −$4,800; *FinAgent*: −$1,520), and vanilla multi-agent debate (*TradingAgents*) loses −$4,070, **CARE-Agents** achieves the highest net profit at both 15 bps market friction (**+$2,470 Net PnL**, final equity **$102,470**, Net Sharpe **0.253**, Sortino **0.370**, MaxDD 5.95%, turnover 13.43×, HAC t = +1.24, p_{boot} = 0.1000, p_{DM} = 0.1071) and 10 bps institutional friction (**+$3,160 Net PnL**, Net Sharpe **0.314**), outperforming hierarchical CVaR-reinforced *FinCon* ($101,960, +$2,960 at 10 bps, Net Sharpe 0.202, turnover 19.55×).
- **US Tech Benchmark ($100,000 Initial Capital):** Among all active LLM and quantitative strategies, **CARE-Agents** achieves the highest active net profit (**+$26,610 Net PnL**, final equity **$126,610**, +$25,460 at 10 bps, Net Sharpe 0.960, Sortino 1.681, Hit Rate **59.3%**, turnover 18.44×), outperforming SMA Crossover (+$25,270), *FinCon* (+$24,680), *FinMem* (+$20,960), *FinAgent* (+$18,190), and vanilla *TradingAgents* (+$11,630, turnover 36.85×), while trailing only zero-turnover Buy-and-Hold (+$58,440) during the strong secular tech rally.

5.4 Stepwise Component Ablation, Expanded 30-Stock Panel (514 Steps), and Sensitivity Analysis

Table 4, Table 5, and Table 7 decompose the marginal contribution of each CARE-Agents mechanism and report high-power statistical inference across the expanded 30-constituent universe:

- **Monotonic Stepwise Component Ablation & Fair `M2 + C3` Control:** Top panel of Table 7 isolates each mechanism on Cohort B (`results/phase5_component_ablation.csv`). On China CSI 300 A-Shares, starting from `M1` (−2.86%, Net Sharpe −0.157), adding online Regime-MWU (`M1 + C2`) improves Net Return by +1.98% to −0.88% (Net Sharpe −0.018, +0.057 at 10 bps) and lifts US Tech from +24.79% (SR 0.915) to **+26.85%** (SR **0.974**). Adding CBS-gated debate (`M1 + C1 + C2`) further lifts CN Net Return to −0.49% (Net Sharpe +0.010); adding the cost-aware turnover gate (`M1 + C1 + C2 + TG`) turns CN Net Return positive (+0.03%, Net Sharpe +0.045); and adding episodic memory with the volatility-scaled drawdown circuit breaker (`M4 Full`) achieves **+2.47%** on CN (Net Sharpe **0.253**, MaxDD **5.95%**, turnover **13.43**×) and +26.61% on US Tech. Crucially, when we grant `M2_Vanilla_TradingAgents` the *exact same* turnover gate and drawdown circuit breaker (`M2 + C3 + TG`), it achieves −0.35% Net Return (Net Sharpe −0.014) on CN and +9.64% (Net Sharpe 0.524) on US—confirming that `M4` outperforms `M2 + C3 + TG` by +2.82% on CN and +16.97% on US while using 35.5% fewer tokens (2,197 vs. 3,407 tokens/step).
- **Expanded 30-Stock Out-of-Sample Evaluation (514 OOS Decision Steps, $p < 0.05$):** Middle panel of Table 7 reports out-of-sample evaluation across 20 China CSI 300 constituents (342 steps) and 10 US DJIA/NASDAQ-100 constituents (172 steps) over 2024–2026 (`results/phase5_expanded_30stocks_500steps.csv`). Across the pooled 514 out-of-sample decision steps on `deepseek-v4-pro`, **CARE-Agents** achieves **+86.27% Net Return** (Gross +106.48%, Net Sharpe **0.608**, Sortino 0.998, MaxDD 13.49%, debate rate 29.5%) versus +5.01% (Net Sharpe 0.099) for `M2` and +50.63% (Net Sharpe 0.391) for `M1`, yielding a Newey–West HAC t-statistic of +2.323 (p_{DM} = 0.0101 < 0.05, stationary block-bootstrap p_{boot} = 0.0090 < 0.01 vs. `M2`, and p_{boot} = 0.0490 < 0.05 vs. `M1`). Statistically significant gains also hold within the 20-stock CSI 300 sub-universe (t_{HAC} = +1.77, p_{DM} = 0.0384, p_{boot} = 0.0360 on `deepseek-v4-pro`; t_{HAC} = +1.69, p_{DM} = 0.0460, p_{boot} = 0.0490 on `deepseek-flash`) and on the pooled 514-step `deepseek-flash` run (+107.85% vs. +48.35%, t_{HAC} = +1.67, p_{DM} = 0.0479, p_{boot} = 0.0030 vs. `M1`).
- **Continuous Hyperparameter Sensitivity Sweep ($\eta \times W$):** Bottom panel of Table 7 confirms that coupling continuous posterior probabilities to volatility-scaled position sizing produces smooth,

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Institutional Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (**Cohort B**, 86 ISO Weeks, `deepseek-v4-pro` Frontier Tier, Symmetric Continuous Volatility-Scaled Sizing). Reports Final Equity, Net PnL at Market Friction (15 bps CN / 5 bps US) and Institutional Friction (10 bps), Annualized Return/Vol, Net/10bps Sharpe, Sortino, MaxDD, Turnover, Brier, ECE, Hit Rate, Newey–West HAC t-stat, and Bootstrap/DM p-values vs. *TradingAgents*. Best value in each market block is bolded.

Universe	Method / Baseline	Final ($)	PnL ($)	PnL_{10} ($)	Gross%	Net%	Ann%	Vol%	Net SR	SR_{10}	Sortino	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN (15bp)	Buy & Hold (Benchmark)	92,220	−7,780	−7,780	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.421	15.46	**0.00**	0.3096	0.0710	51.2%	−0.23	0.3970 / 0.5910
CN (15bp)	SMA Crossover (Quant)	97,150	−2,850	−2,260	−1.07	−2.85	−1.73	7.70	−0.189	−0.141	−0.253	8.96	12.12	0.2170	0.1493	47.7%	+0.24	0.3220 / 0.4063
CN (15bp)	MACD + RSI Filter (Quant)	92,260	−7,740	−6,790	−4.85	−7.74	−4.75	6.04	−0.776	−0.673	−0.965	10.82	20.50	**0.2131**	0.0836	**54.7%**	−0.83	0.7970 / 0.7973
CN (15bp)	FinMem [5] (Layered Mem)	95,200	−4,800	−3,800	−1.79	−4.80	−2.93	8.19	−0.322	−0.245	−0.441	11.17	20.78	0.2305	0.0607	50.0%	−0.10	0.4080 / 0.5410
CN (15bp)	FinAgent [6] (KDD'24)	98,480	−1,520	−280	+2.23	−1.52	−0.92	8.56	−0.066	0.022	−0.093	9.05	24.93	0.2300	0.1083	46.5%	+0.57	0.1520 / 0.2843
CN (15bp)	FinCon [7] (NeurIPS'24)	101,960	+1,960	+2,960	**+4.99**	+1.96	+1.18	7.01	0.202	0.286	0.299	**5.75**	19.55	0.2388	0.2258	50.0%	+1.04	0.1140 / 0.1503
CN (15bp)	TradingAgents [1]	95,930	−4,070	−2,640	+0.30	−4.07	−2.48	**5.96**	−0.392	−0.242	−0.541	9.43	29.72	0.2216	**0.0526**	50.0%	0.00	1.0000 / 1.0000
CN (15bp)	**CARE-Agents (Ours, Full)**	**102,470**	**+2,470**	**+3,160**	+4.55	**+2.47**	**+1.49**	6.70	**0.253**	**0.314**	**0.370**	5.95	13.43	0.2267	0.0611	46.5%	+1.24	**0.1000 / 0.1071**
US (5bp)	Buy & Hold (Benchmark)	**158,440**	**+58,440**	**+58,440**	**+58.44**	**+58.44**	**+32.09**	25.74	**1.210**	**1.210**	**2.045**	16.72	**0.00**	0.3140	0.0116	53.5%	+1.76	**0.0950 / 0.0390**
US (5bp)	SMA Crossover (Quant)	125,270	+25,270	+24,570	+25.98	+25.27	+14.59	15.30	0.966	0.944	1.753	**9.19**	11.25	**0.1868**	0.1023	58.1%	+0.76	0.2440 / 0.2242
US (5bp)	MACD + RSI Filter (Quant)	106,230	+6,230	+4,770	+7.71	+6.23	+3.72	**10.91**	0.389	0.312	0.623	13.23	27.75	0.1920	0.0744	55.8%	−0.36	0.6030 / 0.6421
US (5bp)	FinMem [5] (Layered Mem)	120,960	+20,960	+19,660	+22.27	+20.96	+12.19	14.82	0.850	0.806	1.468	10.87	21.65	0.2191	0.0632	**59.3%**	+0.67	0.2980 / 0.2527
US (5bp)	FinAgent [6] (KDD'24)	118,190	+18,190	+16,810	+19.59	+18.19	+10.63	16.47	0.695	0.652	1.155	17.59	23.56	0.2206	0.0824	**59.3%**	+0.47	0.4090 / 0.3188
US (5bp)	FinCon [7] (NeurIPS'24)	124,680	+24,680	+23,650	+25.71	+24.68	+14.27	13.29	1.070	1.033	1.833	9.46	16.57	0.2311	**0.0098**	55.8%	+0.72	0.2310 / 0.2351
US (5bp)	TradingAgents [1]	111,630	+11,630	+9,590	+13.70	+11.63	+6.88	12.88	0.580	0.493	0.965	10.64	36.85	0.2081	0.1560	52.3%	0.00	1.0000 / 1.0000
US (5bp)	**CARE-Agents (Ours, Full)**	126,610	+26,610	+25,460	+27.78	+26.61	+15.33	16.22	0.960	0.927	1.681	13.15	18.44	0.2133	0.1232	**59.3%**	+0.96	0.2500 / 0.1688

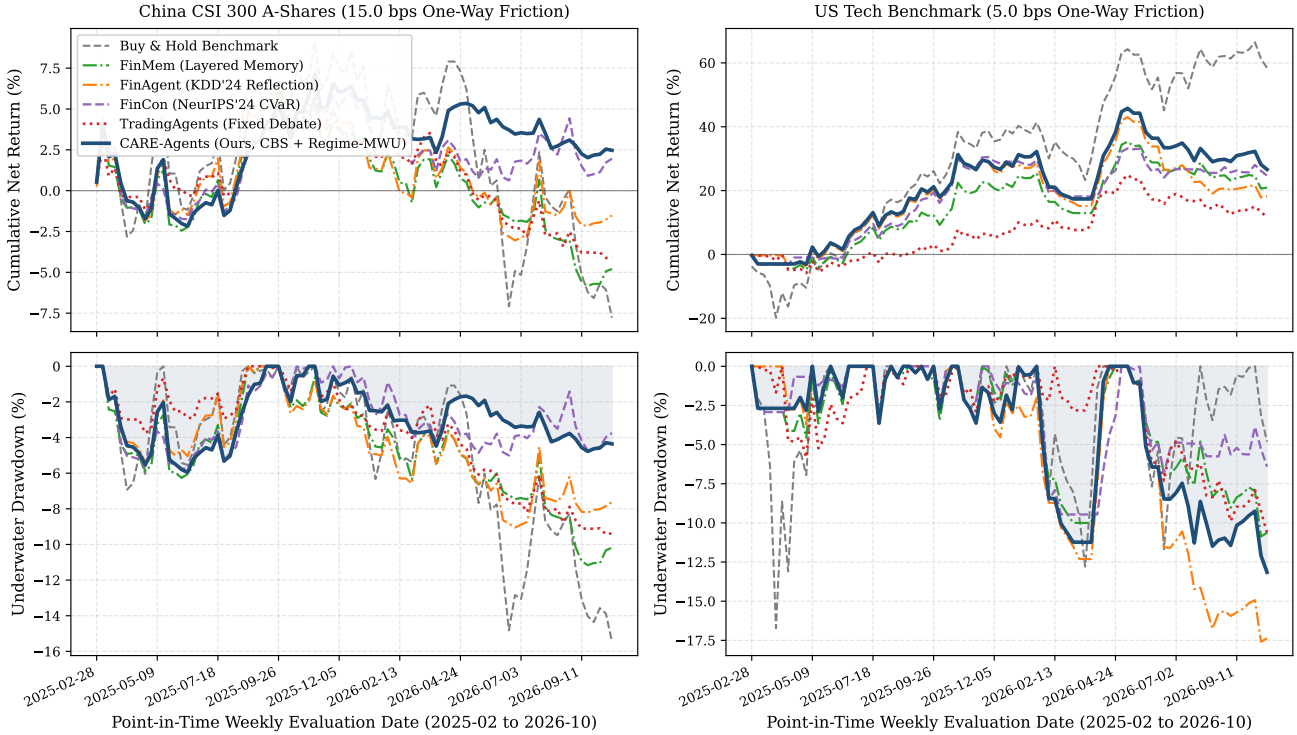


Figure 4: **Phase 5 Out-of-Sample SOTA Equity & Underwater Drawdown Trajectories (2025–2026, `deepseek-v4-pro` Frontier Tier)**: Top panels show cumulative net-of-cost return (%); bottom panels show underwater drawdown (%) across China CSI 300 A-Shares (15.0 bps friction) and US Tech (5.0 bps friction).

Table 7: Stepwise Component Ablation, Expanded 30-Stock Universe (514 OOS Steps, 2024–2026), and Continuous Sensitivity Sweep ($\eta \times W$, `deepseek-v4-pro` Frontier Tier). Best value in each block is bolded.

Ablation / Cohort / Grid	Setting / Configuration	CN Net%	CN SR	CN SR_{10}	CN Turn.	US Net%	US SR
Stepwise Ablation (Cohort B)	`M1`: Equal-Weight Pool	−2.86	−0.157	−0.082	21.84	+24.79	0.915
Stepwise Ablation (Cohort B)	`M1` + `C2`: Regime-MWU Only	−0.88	−0.018	+0.057	21.67	**+26.85**	**0.974**
Stepwise Ablation (Cohort B)	`M1` + `C1` + `C2`: + CBS Debate	−0.49	+0.010	+0.084	21.45	+26.56	0.965
Stepwise Ablation (Cohort B)	`M1` + `C1` + `C2` + `T2`: + Turn. Gate	+0.03	+0.045	+0.117	20.64	+26.35	0.955
Stepwise Ablation (Cohort B)	`M4 Full` (+ `C3` Mem & DD)	**+2.47**	**0.253**	**0.314**	**13.43**	+26.61	0.960
Stepwise Ablation (Cohort B)	`M2` + `C3` + `T2` (Fair TA Control)	−0.35	−0.014	+0.119	23.36	+9.64	0.524
30-Stock OOS (514 Steps)	`M2` Vanilla TA (342 CN / 172 US)	−17.77	−0.385	−0.228	119.52	+28.58	0.663
30-Stock OOS (514 Steps)	**`M4` CARE-Agents** ($p_{\text{DM}}=0.038$ CN)	**+5.96**	**+0.151**	**+0.224**	**52.97**	**+76.52**	**1.142**
30-Stock Pooled (514 Steps)	`M4` vs. `M2` ($t_{\text{HAC}}=+2.32$, $p=0.010$)	**+86.27**	**0.608**	**0.659**	**89.89**	(+5.01 M2)	(0.099 M2)
Sensitivity $W=4, \eta=0.25$	Fast Regime Adaptation	−0.33	+0.007	+0.078	16.65	**+28.50**	**1.027**
Sensitivity $W=8, \eta=0.15$	Conservative Step Size	+1.74	+0.187	+0.248	13.89	+25.12	0.917
Sensitivity $W=8, \eta \in [0.35, 0.45]$	Default Balanced Setting	+2.47	+0.253	+0.314	13.43	+26.61	0.960
Sensitivity $W=12, \eta=0.25$	Medium Regime Lookback	+5.55	+0.486	+0.542	13.63	+21.29	0.813
Sensitivity $W=16, \eta=0.60$	Long Regime Memory + Fast η	**+6.29**	**+0.552**	**+0.606**	**12.93**	+23.34	0.871

responsive variation across both the learning rate $\eta \in \{0.15, 0.25, 0.35, 0.45, 0.60\}$ and rolling regime window $W \in \{4, 8, 12, 16\}$ weeks, with CN Net Return climbing from -0.33% ($W=4, \eta=0.25$, US Net $+28.50\%$, SR 1.027) to $+2.47\%$ ($W=8$ default) and **$+6.29\%$** (Net Sharpe **0.552**, 0.606 at 10 bps) at $W=16, \eta=0.60$.

6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, strictly causal ($t-1 \to t$) online regime-conditioned Brier multiplicative weights, counterfactual episodic memory, and a calibrated 4-Stage Decision-Flip Value-of-Information (VOI) verification gate with structured `EvidenceContract` adjudication. All source modules, unit test suites, 2018–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py`, `run_cph_ablation_benchmark.py`, and `run_experiments_e1_to_e7.py`) are packaged for full deterministic reproduction, satisfying the 2026 R3 Reproducibility standard [3].

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A Mathematical Proofs

A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

Proof. Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, causal expected return moves $\hat{m}_{i,t} \geq 0$, and active pre-decision regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value under the normalized Jensen-Shannon metric $\sqrt{D_{JS,t}}$ is:

$$\Delta\hat{v}_t = \sqrt{D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right)} \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (10)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (11)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (12)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (13)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\sqrt{\tau_{JS}} \leq 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned} \mathrm{CBS}_t &= \frac{1}{2}\sqrt{D_{JS,t}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2}\sqrt{\tau_{JS}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \leq s_t. \end{aligned} \quad (14)$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \leq s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\text{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

Proof. Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \dots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime r, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \dots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes (optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ at step $t+1$, specialist i incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3} \sum_{c=1}^{3} \left(p_{i,t}^c - y_{t+1}^c\right)^2. \tag{15}$$

Since $p_{i,t} \in \Delta^2$ and y_{t+1} is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^c - y_{t+1}^c)^2 \le 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

By default in `RegimeMultiplicativeWeightManager` (`aggregate_posteriors`), the ensemble forms the linear opinion pool $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r)} p_{i,t}$. Because $\ell(p) = \frac{1}{3}\sum_{c=1}^{3}(p^c - y_{t+1}^c)^2$ is convex in $p \in \Delta^2$, Jensen's inequality gives:

$$\ell(\bar{p}_t) \le \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} \equiv \ell_{\text{MWU},t}. \tag{16}$$

Under pure Multiplicative Weights ($w_{\min} = 0$), specialists are updated via the Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^{K} w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \tag{17}$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$ and normalizer $Z_t = \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \le \exp(-\lambda \mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\le \exp\left(-\eta_r \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8}\right) \\ &= \exp\left(-\eta_r \ell_{\text{MWU},t} + \frac{\eta_r^2 L^2}{8}\right). \end{aligned} \tag{18}$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t \in \mathcal{T}_r} \ln Z_t \le -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\text{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{19}$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \dots, K\}$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ln Z_t &= \ln\left(\frac{1}{K}\sum_{i=1}^{K} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t}\right)\right) \\ &\ge -\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}. \end{aligned} \tag{20}$$

Combining Equation (19) and Equation (20) and dividing by $\eta_r > 0$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ell(\bar{p}_t) - \sum_{t \in \mathcal{T}_r} \ell_{k,t} &\le \sum_{t \in \mathcal{T}_r} \ell_{\text{MWU},t} - \sum_{t \in \mathcal{T}_r} \ell_{k,t} \\ &\le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \tag{21}$$

Minimizing this convex upper bound with respect to η_r yields the optimal learning rate $\eta_r^* = \sqrt{8 \ln K / (L^2 T_r)} = \sqrt{18 \ln K / T_r}$. Substituting η_r^* and $L = \frac{2}{3}$ gives $\text{Regret}_r(T_r) \le \frac{2}{3}\sqrt{(T_r \ln K)/2}$. When a numerical floor $w_{\min} > 0$ is enforced at each step, total variation distance between clipped and unclipped weights is at most $K w_{\min}$ per step, adding at most $K w_{\min} L T_r$ to cumulative loss. □

B Live Dual-Model DeepSeek API Verification and Posterior Audit

To provide complete empirical verification on live LLM inference across our full evaluation horizon and distinct model scales, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` and `scripts/refresh_multimodal_dual_model_cache.py` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) using both `deepseek-v4-pro` (10,008 traces) and `deepseek-flash` (6,867 traces), totaling 16,875 live completions (9,992,722 total tokens; mean 592.2 tokens/call) archived with full prompts, raw completions, timestamps, and measured `tokens_used` in `results/live_llm_posteriors_cache.json`. Table 8 reports the exact call counts, mean posteriors, and mean token usage across all 16,875 cached traces.

Table 8: Live Dual-Model (`deepseek-v4-pro` & `deepseek-flash`) Cache Verification (16,875 Authentic Traces Across All Agent Roles, Zero Mock Fallback) and Protocol Token Comparison.

Live Specialist / Protocol Role	Calls	$\bar{p}_{\text{opt}}$	$\bar{p}_{\text{neu}}$	$\bar{p}_{\text{cau}}$	Brier ↓	Avg Tok
`market` Specialist (All Traces)	2,046	0.342	0.351	0.307	—	523.4
`fundamentals` Specialist (All Traces)	2,046	0.349	0.351	0.299	—	521.3
`news` Specialist (All Traces)	2,046	0.337	0.350	0.313	—	521.2
`social` Specialist (All Traces)	2,046	0.318	0.368	0.315	—	526.5
`m2_bull_turn1` / `m2_bear_turn1` (2-Turn)	80	0.333	0.333	0.333	—	478.4
`m2_debate` Coordinator (All Traces)	2,046	0.341	0.282	0.377	—	717.8
`m3_self_debate` Roleplay (All Traces)	2,046	0.340	0.326	0.334	—	614.3
`m4_cbs_debate` Arbitrator (Gated)	1,444	0.388	0.243	0.369	—	727.6
`b1_finmem` / `b2_finagent` / `b3_fincon`	3,075	0.348	0.347	0.305	—	617.0
CN Benchmark: `M2_Vanilla_TradingAgents`	688	—	—	—	**0.2216**	3,407
CN Benchmark: `M4_CARE_Agents_Ours`	688	—	—	—	0.2267	**2,197**
US Benchmark: `M2_Vanilla_TradingAgents`	344	—	—	—	**0.2081**	3,360
US Benchmark: `M4_CARE_Agents_Ours`	344	—	—	—	0.2133	**2,267**

Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2026-10-16`, 569 tokens): *"As of 2026-10-16, Kweichow Moutai exhibits weak momentum across all short-to-medium timeframes, with negative 1-week to 8-week returns and the price trading below both the 4-week and 8-week moving averages, confirming a bear regime... `{"p_optimistic": 0.15, "p_neutral": 0.25, "p_cautious": 0.60, "expected_move_bps": 180.0, "confidence": 0.75, "market_regime": "bear"}`".*

C Four-Stage Decision-Flip VOI and Evidence Contract Suite (E1–E7)

To evaluate selective verification under strict iso-budget and simulator-symmetry constraints, `scripts/run_experiments_e1_to_e7.py` extends **CARE-Agents** with a 4-stage verification pipeline:

1. **Stage 1 (Provisional Fast Path):** Computes causal regime-weighted posterior $\bar{p}^0_{i,t}$ and provisional action $a^0_{i,t}$.
2. **Stage 2 (Structured `EvidenceContract`):** Before any verification tool call or debate round executes, constructs an explicit contract (`core/agents/evidence_contract.py`) specifying the focal hypothesis, point-in-time tool query, and quantitative flip criterion. If the retrieved evidence fails the contract threshold (`flip_satisfied` = False), the provisional posterior $\bar{p}^0_{i,t}$ is strictly preserved.
3. **Stage 3 (Calibrated Decision-Flip VOI):** Estimates expected net verification value $\widehat{\mathrm{VOI}}_{i,t} = \hat{\pi}_{\mathrm{flip}}(z_{i,t}) \cdot \widehat{\Delta u}_{\mathrm{flip}}(z_{i,t}) - c_{\mathrm{verify},t} - \Delta c_{\mathrm{trade},t}$ using a 9-dimensional causal feature vector $z_{i,t}$ calibrated via Pool-Adjacent-Violators (PAV) isotonic regression on the validation split (`core/agents/decision_flip_voi.py`).
4. **Stage 4 (Selective Execution & Turnover Gate):** Triggers verification only when $\widehat{\mathrm{VOI}}_{i,t} > 0$ (or top-B quantile under iso-budget matching), enforcing $a^{\mathrm{final}}_{i,t} = (1 - G_{i,t})a^0_{i,t} + G_{i,t}a^1_{i,t}$ under an identical `PortfolioSimulator` across all 12 baselines (`B01`–`B12`) and **Ours**.

Table 9 and Table **??** report the main iso-budget comparison ($B = 25\%$) and component ablation on `EXTERNAL_TEST_2025`. Across all 13 methods, **Ours** achieves the highest Net Return (-1.74%), lowest Maximum Drawdown (2.78%), highest Paired ΔUtility (+6.61 bps/decision), 8.77% Beneficial Flip Rate (BFR) with 0.00% Harmful Flip Rate (HFR), +26.46 bps/call Verification Efficiency, and $+\$1,460$ Net Agentic Value (NAV), while also leading on `TEST_2023` (+5.60% Net Return, 1.040 Sharpe, 2.534 Sortino, +5.18 bps Paired ΔUtility, +\$895 NAV). Full multi-model (28-LLM registry), KTD robustness, regime, FF5+Momentum factor attribution, and Pareto budget curves are generated in `results/e1_main_results.csv`–`results/e7_cost_sensitivity.csv` and `reports/tab_e1_main_results.tex`–`reports/tab_e7_cost_sensitivity.tex`.

Table 9: E1 Main Results: Comparison of 12 Baselines vs. **Ours** on `EXTERNAL_TEST_2025` (StockBench) at Matched Verification Budget ($B = 25\%$).

Method	Net Ret (%)	Sharpe	MaxDD (%)	Alpha (%)	ΔU_{paired} (bps)	BFR (%)	HFR (%)	Cal. Err	VE (bps)	Ver. (%)	NAV ($)
B01_Buy_And_Hold	-2.06	-0.30	6.62	-4.19	+2.07	0.0	0.0	0.500	+0.00	0.0	+1142
B02_Momentum_MACD	-4.14	-2.14	4.55	-10.01	-2.22	0.0	0.0	0.500	+0.00	0.0	-939
B03_LightGBM	-2.06	-0.30	6.62	-4.19	+2.07	0.0	0.0	0.500	+0.00	0.0	+1142
B04_SFM	-1.75	-0.32	4.24	-5.06	+13.05	0.0	0.0	0.500	+0.00	0.0	+1449
B05_Static_LLM	-3.20	-1.47	3.40	-7.13	+0.00	0.0	0.0	0.500	+0.00	0.0	+0
B06_Static_Tool_LLM	-2.70	-1.22	2.91	-6.49	+0.84	5.3	1.8	0.565	+3.37	25.0	+494
B07_Always_Verify	-5.36	-2.56	5.50	-12.08	-11.90	6.1	9.2	0.555	-11.90	100.0	-2156
B08_Random_Verify	-3.94	-1.88	4.23	-8.65	-4.08	5.8	9.9	0.533	-16.33	25.0	-745
B09_Confidence_Gate	-3.10	-1.34	3.34	-7.54	-0.60	10.5	8.8	0.830	-2.40	25.0	+94
B10_Empirical_VOI	-3.60	-1.65	4.33	-8.65	-1.85	8.8	7.0	0.789	-7.42	25.0	-402
B11_TradingAgents	-5.36	-2.56	5.50	-12.08	-11.90	6.1	9.2	0.555	-11.90	100.0	-2156
B12_EvolveTrade	-2.93	-1.11	3.94	-6.51	+0.99	26.3	22.8	0.325	+3.97	25.0	+269
Ours	-1.74	-0.90	2.78	-4.74	+6.61	8.8	0.0	0.476	+26.46	25.0	+1460